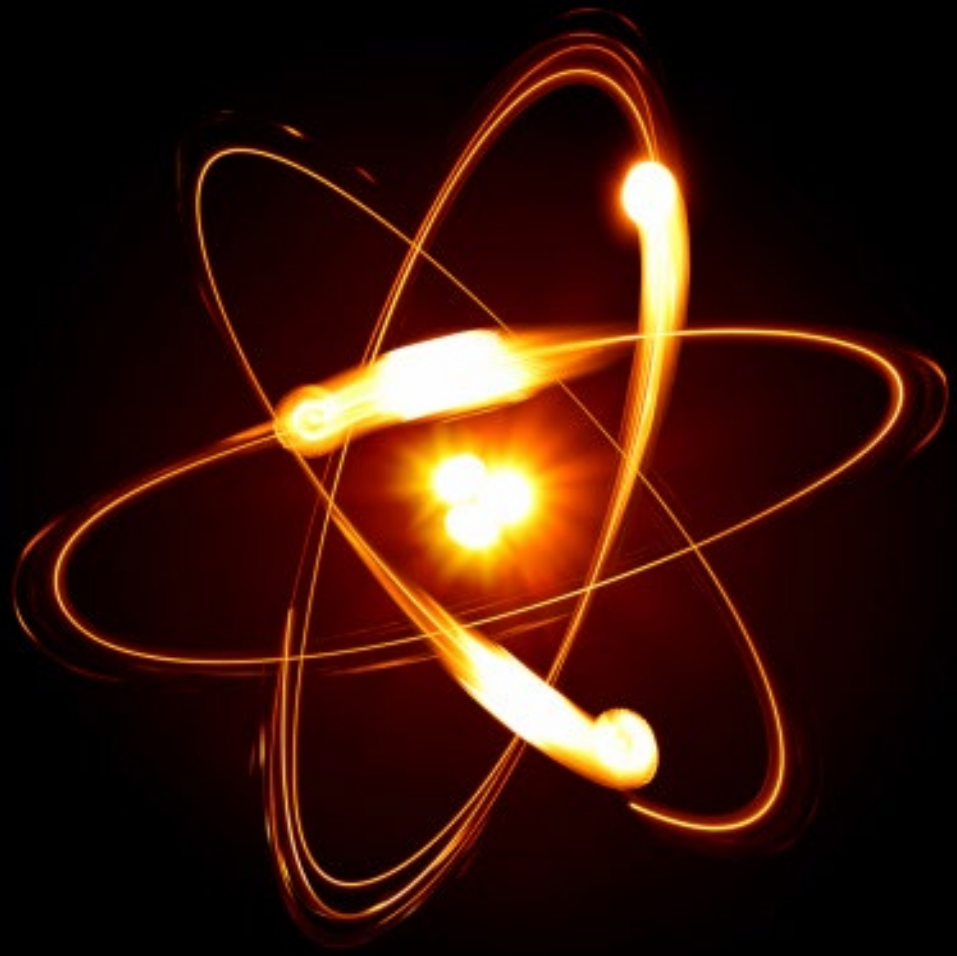
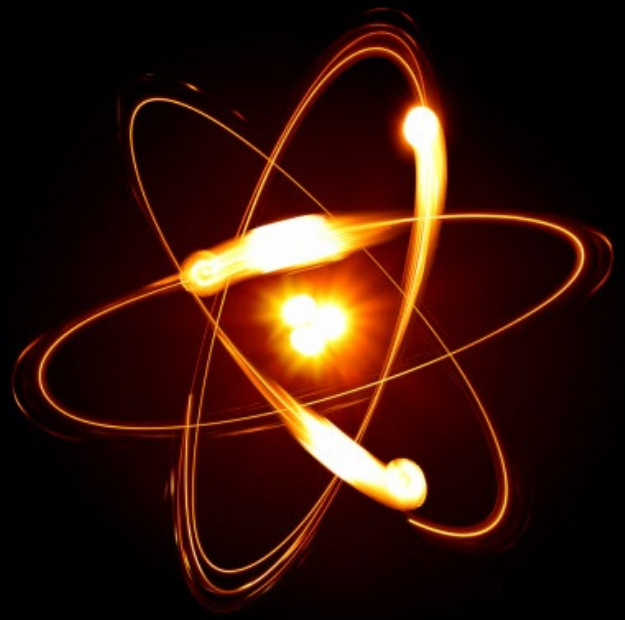
Chapter 23.

Identical particles



Beatriz González del Río

Quantum Physics



Chapter 23.

Identical particles

- What are they?
- Why do they cause problems? Examples
- Solution: symmetrization postulate
- Method to (anti)symmetrize a state
- Measurements for 2 identical particles
- N independent identical particles

Identical particles

- When do we say that two particles are identical?

  When they have the same intrinsic properties: mass, charge, spin.....(no experiment can distinguish one from the other)

- These systems of identical particles have one additional symmetry, namely under the exchange of the identities of two particles. (i.e. there is no change in the properties or evolution of the system if the roles of the two particles are exchanged)

- How are they treated in classical mechanics? They pose no particular problem

  Each particle moves along a **well defined trajectory** which enables to distinguish it from the others.

  Once its state (position and momentum) is specified at an initial time, we can determine its trajectory and predict its state at a later time.

Identical particles in classical mechanics

Once its state (position and momentum) is specified at an initial time, we can determine its trajectory and predict its state at a later time.

We can follow each particle step by step along its trajectory (arrows in the Figure)

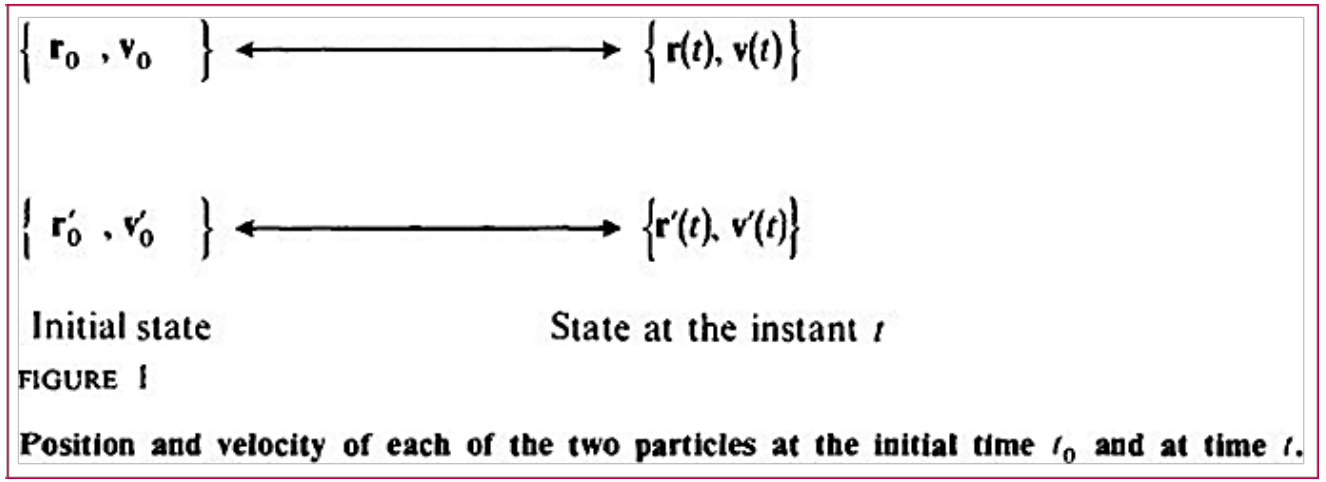


FIGURE 1

Position and velocity of each of the two particles at the initial time t_0 and at time t.

Consequently we can determine the locations of particle (1) and particle (2) at any time.

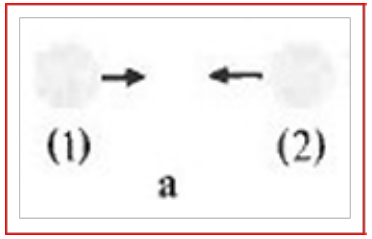
Identical particles in quantum mechanics

Completely different situation in quantum mechanics

- The particles no longer have **definite trajectories**
- Even if at an initial time, their respective wave packets are completely separated, their subsequent evolution may mix them.
- Then, we loose track of the particles
- For example, if in some region the probability of finding two identical particles is not zero, then **it is not possible to determine through an experiment which one has been detected.**
- Consequently if we detect a particle, we do not know if it was, initially, particle 1 or particle 2.

Collision between two identical particles in the CM system

Consider a collision between two identical particles in the CM reference system





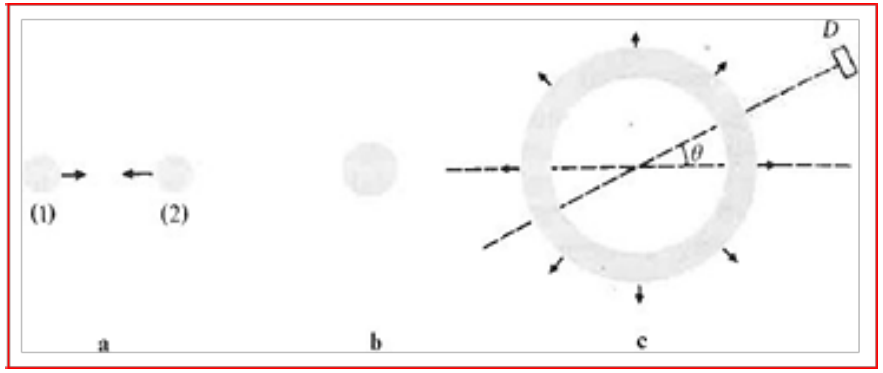
Before the collision (a):

- Initially they are very far from each other.
- We have two completely separated wave packets, directed towards each other.
- We denote as (1) the particle on the left and as (2) the particle on the right.

During the collision (b):

- The two wave packets overlap

Collision between two identical particles in the CM system

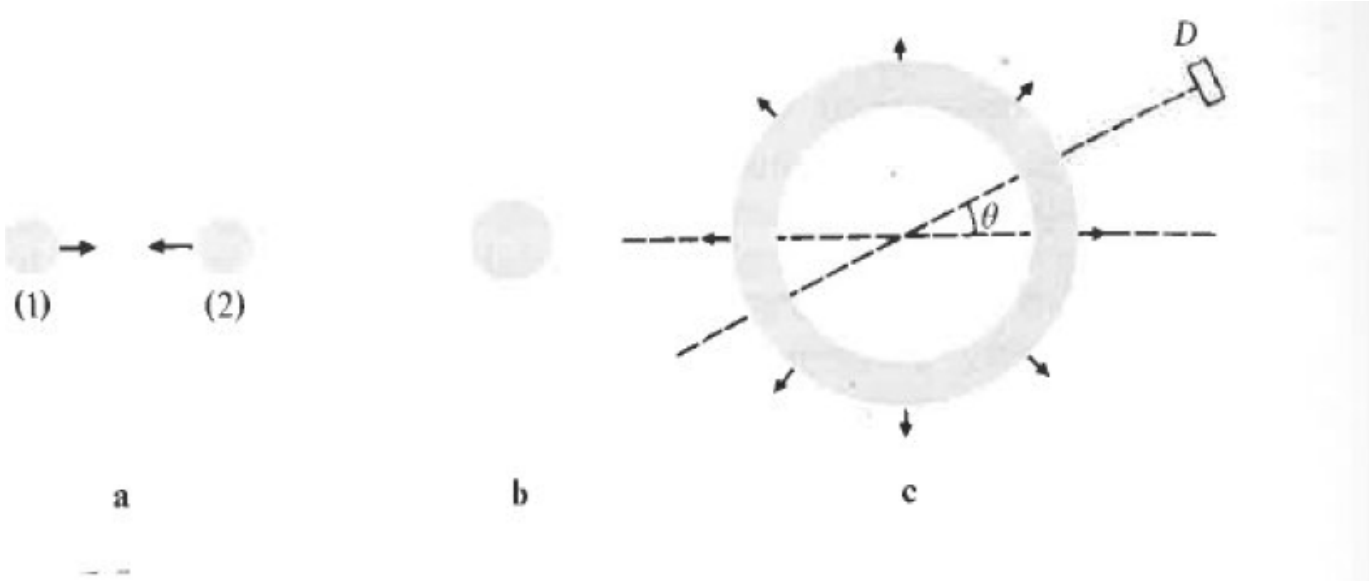


After the collision (c):

- The region of space where the two-particle probability density is non-zero is a spherical shell whose radius increases with time.
- A detector D is placed at an angle θ with the initial direction of wave packet (1).
- D detects a particle. Then we know that the other particle is moving away in the opposite direction due to momentum conservation.
- Which particle has been detected? Particle (1) or particle (2) ?

Collision between two identical particles in the CM system

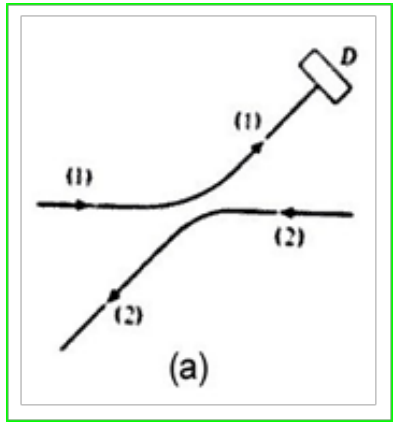
Summary

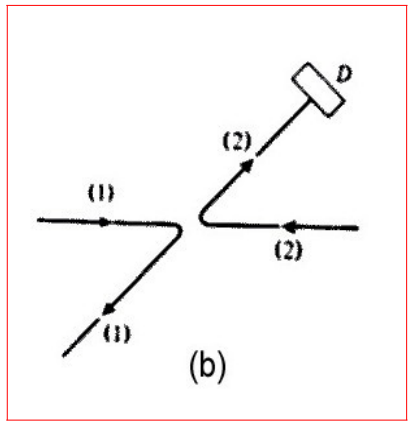


Collision between two identical particles in the center of mass frame : schematic representation of the probability density of the two particles.
Before the collision (fig. a), the two wave packets are clearly separated and can be labeled. During the collision (fig. b), the two wave packets overlap. After the collision (fig. c), the probability density is non-zero in a region shaped like a spherical shell whose radius increases over time. Because the two particles are identical, it is impossible, when a particle is detected at D, to know with which wave packet, (1) or (2), it was associated before the collision.

Collision between two identical particles in the CM system

There are 2 possibilities to describe the measured state





Which is the correct one? (a),(b), some linear combination of them?

It cannot be decided!

What about the probability of the measured state?

There are two possibilities (that is, two distinct state vectors) or final state kets, that describe the measured state:

Should we calculate the probability of path (a) OR path (b) OR both?

Another example

Summary

In the previous example, we considered two wave packets which initially did not overlap, so we could label them.

But when we tried to describe (mathematically) the result of a position measurement associated to the final state, we ran into difficulties (there were two different kets associated to a single physical state).

In fact, similar problems may appear when trying to describe an initial state.

This gives rise to the so-called **exchange degeneracy**

Another example

System composed of two identical particles of spin 1/2.

We measure the *z* component of the spins and obtain that one has $\hbar/2$ and the other $-\hbar/2$.

As usual, $\vec{S}_1$ and $\vec{S}_2$ denote the two spin observables and $\{|\varepsilon_1, \varepsilon_2\rangle\}$ is the associated usual orthonormal basis associated with the total state space $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2$.

In this total state space $\mathcal{E}$, there are two mathematical states that describe this (same) physical situation, namely, $|+,-\rangle$ and $|-,+\rangle$.

These two kets generate a 2-dimensional subspace whose vectors are of the form

$$|\Phi\rangle = \alpha|+,-\rangle + \beta|-,+\rangle \qquad \text{with} \qquad |\alpha|^2 + |\beta|^2 = 1$$

All these mathematical kets represent the same physical state. This is called

EXCHANGE DEGENERACY

The key point is that a complete set of measurements does not determine a unique state vector!

Another example: Further complications

This exchange degeneracy creates fundamental difficulties.

Suppose that, in the previous state $|\Phi\rangle$, we now measure S_{1x} and S_{2x}. What is the probability of finding $\hbar/2$ and $\hbar/2$?

Remember that the considered eigenstate of S_x is: $|+\rangle_x = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$

and therefore

$$|+,+\rangle_x = |+\rangle_x \otimes |+\rangle_x = \frac{1}{2}(|+,+\rangle + |+,-\rangle + |-,+\rangle + |-,-\rangle)$$

Then

$$\mathcal{P}(\uparrow_x,\uparrow_x) = \left|{}_x\langle +,+|\Phi\rangle\right|^2 = \left|\frac{1}{2}(\alpha+\beta)\right|^2$$

The different possible states $|\Phi\rangle$ give different results !!!

It'll be necessary to add a new postulate to those we already know.

Preamble to Postulate 6 (symmetrization postulate)

Those problems related to the exchange degeneracy appear in ALL systems with N > 1 identical particles.

System of three identical particles. Each particle has its own state space and observables. So, we have $\mathcal{E}_1$, $\mathcal{E}_2$ and $\mathcal{E}_3$ and for the whole three particle system $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2 \otimes \mathcal{E}_3$

Let B(1) an observable in $\mathcal{E}_1$ (assume it is a CSCO)

Spectrum $\{b_n; n = 1,2,3, \dots\}$ Basis $\{|1, b_n\rangle, n = 1,2,3, \dots\}$

Obviously

B(2) an observable in $\mathcal{E}_2$ (it is a CSCO)

B(3) an observable in $\mathcal{E}_3$ (it is a CSCO)

Orthonormal basis in $\mathcal{E}$ is $\{|1, b_i; 2, b_j; 3, b_k\rangle \; i, j, k = 1,2, \dots\}$

The $|1, b_i; 2, b_j; 3, b_k\rangle$ are common eigenvectors of B(1), B(2) and B(3)

The three particles are identical. So, we can NOT measure B(1), B(2) or B(3) **BUT** the magnitude B for each particle.

Preamble to Postulate 6 (symmetrization postulate)

The three particles are identical. So, we can not measure B(1), B(2) or B(3) **BUT** the magnitude B for each particle.

Suppose we have measured and found the result b_n, b_p and b_q

State of the system after the measurement ???

Any ket of the subspace spanned by the eigenvectors

$$|1, b_n; 2, b_p; 3, b_q\rangle \qquad |1, b_q; 2, b_n; 3, b_p\rangle \qquad |1, b_p; 2, b_q; 3, b_n\rangle$$

$$|1, b_n; 2, b_q; 3, b_p\rangle \qquad |1, b_p; 2, b_n; 3, b_q\rangle \qquad |1, b_q; 2, b_p; 3, b_n\rangle$$

The exchange degeneracy attacks AGAIN !

Conclusion. A complete measurement on each of the particles DOES NOT allow the determination of a UNIQUE ket of the $\mathcal{E}$.

Preamble to Postulate 6 (permutation operator)

Introduce in $\mathcal{E}$ some operators that permute the various particles of the system

System of **two particles** with same spin.

There is no need to be identical. But the same state space (isomorphic)

Particle 1 with state space $\mathcal{E}_1$ and basis $\{|u_i\rangle,\ i = 1,2,3,\dots\}$

Particle 2 with state space $\mathcal{E}_2$ and similar basis $\{|u_i\rangle,\ i = 1,2,3,\dots\}$

For the two-particle system, we have $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2$ and the basis:

$$\left\{\left|1{:}\,u_i; 2{:}\,u_j\right\rangle \ \ i,j = 1,2,\dots\right\}$$

Note that: $\left|1{:}\,u_i; 2{:}\,u_j\right\rangle$ is different from $\left|1{:}\,u_j; 2{:}\,u_i\right\rangle$ (if $i \neq j$).

We define the permutation operator, P_{21}, acting as

$$P_{21}\left|1{:}\,u_i; 2{:}\,u_j\right\rangle = \left|1{:}\,u_j; 2{:}\,u_i\right\rangle$$

Its action on any ket can be obtained by expanding the ket on the previous basis.

Preamble to Postulate 6 (permutation operator)

Symmetric and Antisymmetric kets

Properties of P_{21}

- $P_{21}P_{21} = \mathbb{I} \quad \Longrightarrow \quad P_{21}^{-1} = P_{21}$
- $P_{21}^{\dagger} = P_{21} \quad \Longrightarrow \quad P_{21}$ is hermitian

The matrix elements of P_{21} in the $\{ \mid 1 : u_i ; 2 : u_j \rangle \}$ basis are:

$$\langle 1 : u_{i'} ; 2 : u_{j'} \mid P_{21} \mid 1 : u_i ; 2 : u_j \rangle = \langle 1 : u_{i'} ; 2 : u_{j'} \mid 1 : u_j ; 2 : u_i \rangle = \delta_{i'j} \, \delta_{j'i}$$

$$\langle 1 : u_{i'} ; 2 : u_{j'} \mid P_{21}^{\dagger} \mid 1 : u_i ; 2 : u_j \rangle = (\langle 1 : u_i ; 2 : u_j \mid P_{21} \mid 1 : u_{i'} ; 2 : u_{j'} \rangle)^*$$
$$= (\langle 1 : u_i ; 2 : u_j \mid 1 : u_{j'} ; 2 : u_{i'} \rangle)^* = \delta_{ij'} \, \delta_{ji'}$$

- Its eigenvalues are real. Moreover, those are: ± 1.
- Eigenvectors of P_{21} with eigenvalue +1 are called symmetric $\quad P_{21}|\psi_s\rangle = |\psi_s\rangle \quad \Longrightarrow |\psi_s\rangle$ symmetric
- Eigenvectors of P_{21} with eigenvalue -1 are called anti-symmetric $\quad P_{21}|\psi_A\rangle = -|\psi_A\rangle \quad \Longrightarrow |\psi_A\rangle$ anti-symmetric

Preamble to Postulate 6 (symmetrization postulate)

Symmetrizer and Antisymmetrizer

- Define new operators $S = \frac{1}{2}(\mathbb{I} + P_{21})$ and $\mathrm{A} = \frac{1}{2}(\mathbb{I} - P_{21})$
- They are projectors $(S^2 = S,\ A^2 = A)$ and Hermitian $\left(S^\dagger = S,\ A^\dagger = A\right)$
- $SA = AS = 0$ $\qquad S + A = \mathbb{I}$
- If $|\Psi\rangle$ is an arbitrary ket, then:

  $S|\Psi\rangle$ is a symmetric ket $\qquad P_{21}S|\Psi\rangle = S|\Psi\rangle$

  $\mathrm{A}|\Psi\rangle$ is an anti-symmetric ket. $\qquad P_{21}A|\Psi\rangle = -A|\Psi\rangle$

- S and A are called the symmetrizer and anti-symmetrizer operators
- $[P_{21}, S] = [P_{21}, A] = 0$
- $SP_{21} = P_{21}S = S$ $\qquad AP_{21} = P_{21}A = -A$
- $SP_{21}|\Psi\rangle = S|\Psi\rangle$ $\qquad AP_{21}|\Psi\rangle = -A|\Psi\rangle$

Preamble to Postulate 6 (symmetrization postulate)

System of three particles (not necessarily identical) with same spin. State spaces: $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$.

Now, for the whole system: $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2 \otimes \mathcal{E}_3$.

Basis: $\{|1, u_i; 2, u_j; 3, u_k\rangle \quad (i, j, k = 1,2, \dots)\}$

Now, we can define 3! permutation operators: $P_{123}, P_{132}, P_{231}, P_{213}, P_{312}, P_{321}$

Its action is defined as: $P_{231}|1, u_i; 2, u_j; 3, u_k\rangle = |2, u_i; 3, u_j; 1, u_k\rangle = |1, u_k; 2, u_i; 3, u_j\rangle$

Note that P_{123} coincides with the identity operator.

The permutation operators do not commute with each other.

System N particles implies N! permutation operators.

Preamble to Postulate 6 (symmetrization postulate)

Completely symmetric or antisymmetric kets.

System of N particles with same spin

We can define N! permutation operators. Let P_α denote an arbitrary permutation operator associated with this system of N particles.

There exist certain kets which are simultaneously eigenvectors of **ALL** the permutation operators.

A ket $|\psi_S\rangle$ such that $P_\alpha|\psi_S\rangle = |\psi_S\rangle$. Completely Symmetric ket

A ket $|\psi_A\rangle$ such that $P_\alpha|\psi_A\rangle = \epsilon_\alpha|\psi_A\rangle$. Completely Antisymmetric ket

$\epsilon_\alpha = 1\ (-1)$ if P_α is even (odd) permutation.

Now,

$\mathcal{E}_S$= subspace comprising all completely symmetric kets

$\mathcal{E}_A$= subspace comprising all completely antisymmetric kets

Preamble to Postulate 6 (symmetrization postulate)

Symmetrizer and Antisymmetrizer

Consider the operators $S = \frac{1}{N!}\sum_{\alpha} P_\alpha$ $A = \frac{1}{N!}\sum_{\alpha} \epsilon_\alpha P_\alpha$

Given any arbitrary permutation, P_α, we have

$P_\alpha S = S P_\alpha = S$ $P_\alpha A = A P_\alpha = \epsilon_\alpha A$

Therefore, for any ket $|\psi\rangle \in \mathcal{E}$ we have

$P_\alpha S|\psi\rangle = S|\psi\rangle$ $P_\alpha A|\psi\rangle = \epsilon_\alpha A|\psi\rangle$

- S is the symmetrizer operator and A is the antisymmetrizer operator
- They are hermitian, namely $S^\dagger = S$ $A^\dagger = A$
- They are projectors (onto $\mathcal{E}_S$ and $\mathcal{E}_A$ respectively), so $S^2 = S$ and $A^2 = A$
- $AS = SA = 0$ But now, $S + A \neq \mathbb{I}$.

Postulate 6 (symmetrization postulate)

When a system is made of several identical particles, **only certain kets** of the state space represent physically acceptable states (PAS):

- The PAS are either completely symmetric or completely antisymmetric kets.
- Those particles for which the physical kets are completely antisymmetric are called fermions
- Those particles for which the physical kets are completely symmetric are called bosons

The postulate reduces the dimension of the acceptable state space, from the complete tensor product, $\mathcal{E}$, to either $\mathcal{E}_S$ or $\mathcal{E}_A$.

Moreover, there is experimental evidence that:

- The particles with half-integer spin $\left(s = \frac{1}{2}, \frac{3}{2}, \ldots\right)$ are fermions.
- The particles with integer spin $(s = 0,1,2, \ldots)$ are bosons.

Postulate 6 (symmetrization postulate)

Removing the exchange degeneracy

Let $|u\rangle$ be a ket representing a physical state of a system of N identical particles

We know that for any permutation operator, P_α, the new ket $P_\alpha|u\rangle$ represents the SAME physical state.

Moreover, the same happens for any ket belonging to the subspace ε_u generated by $|u\rangle$ and all its permutations $P_\alpha|u\rangle$.

So we have exchange degeneracy.

But the new postulate restricts the class of kets that can be used to describe the physical state: they must belong to $\mathcal{E}_S$ or $\mathcal{E}_A$

So, ε_u MUST contain a SINGLE ket of $\mathcal{E}_S$ or $\mathcal{E}_A$. This is so because

$$S|u\rangle = SP_\alpha|u\rangle \qquad\qquad A|u\rangle = \epsilon_\alpha AP_\alpha|u\rangle$$

which means that the projections onto $\mathcal{E}_S$ or $\mathcal{E}_A$ of all the kets belonging to ε_u , are collinear.

How can a symmetric or antisymmetric state be constructed?

Summary:

Out of all the kets which span ε_u , there is only ONE ket which must be associated with the physical state considered:

- S $|u\rangle$ for bosons
- A $|u\rangle$ for fermions

Rule for the construction of the UNIQUE ket (the physical ket) corresponding to a given physical state of N identical particles.

(1) Number the particles arbitrarily and construct the ket $|u\rangle$ corresponding to the physical state considered.

(2) Apply S or A to $|u\rangle$ depending on whether the identical particles are bosons or fermions.

(3) Normalize the ket so obtained.

How can a symmetric or antisymmetric state be constructed? Example

Consider a system of two identical particles: one is in the state $|\varphi\rangle$ and the other in the state $|\chi\rangle$ (orthonormalized)

We label particles: $|1,\varphi\rangle$ and $|2,\chi\rangle \implies |u\rangle = |1,\varphi;2,\chi\rangle$

Bosons: Symmetrize $S|u\rangle = (1/2)[|1,\varphi;2,\chi\rangle + |1,\chi;2,\varphi\rangle]$

Fermions: Antisymmetrize $A|u\rangle = (1/2)[|1,\varphi;2,\chi\rangle - |1,\chi;2,\varphi\rangle]$

Normalize: Replace $\frac{1}{2}$ by $\frac{1}{\sqrt{2}}$

What happens if $|\varphi\rangle = |\chi\rangle$? $\implies |u\rangle = |1,\varphi;2,\varphi\rangle$

For bosons, we have $S|u\rangle = |u\rangle$

For fermions, we have $A|u\rangle = 0$ $\Rightarrow$ there is no antisymmetric ket to describe this kind of state $\Rightarrow$ **two identical fermions cannot be in the same individual state** (Pauli's exclusion principle.)

How can a symmetric or antisymmetric state be constructed? Example

Consider a system of three identical particles in states $|\varphi\rangle$, $|\chi\rangle$ and $|\omega\rangle$ (orthonormalized)

We construct $|u\rangle = |1,\varphi;2,\chi;3,\omega\rangle$

Bosons

$$S|u\rangle = (1/3!)\sum_{\alpha} P_\alpha |u\rangle =$$

$$= \frac{1}{6}[|1,\varphi;2,\chi;3,\omega\rangle + |1,\omega;2,\varphi;3,\chi\rangle + |1,\chi;2,\omega;3,\varphi\rangle + |1,\varphi;2,\omega;3,\chi\rangle + |1,\chi;2,\varphi;3,\omega\rangle + |1,\omega;2,\chi;3,\varphi\rangle]$$

Normalize: Replace $\frac{1}{6}$ by $\frac{1}{\sqrt{6}}$

- What if $|\varphi\rangle = |\chi\rangle$ which implies $|u\rangle = |1,\varphi;2,\varphi;3,\omega\rangle$

$$S|u\rangle = \frac{1}{3}[|1,\varphi;2,\varphi;3,\omega\rangle + |1,\varphi;2,\omega;3,\varphi\rangle + |1,\omega;2,\varphi;3,\varphi\rangle]$$

Normalize: Replace $\frac{1}{3}$ by $\frac{1}{\sqrt{3}}$

- What if $|\varphi\rangle = |\chi\rangle = |\omega\rangle$ which implies $|u\rangle = |1,\varphi;2,\varphi;3,\varphi\rangle$ $\quad S|u\rangle = |1,\varphi;2,\varphi;3,\varphi\rangle$

How can a symmetric or antisymmetric state be constructed? Example

Consider a system of three identical particles in states $|\varphi\rangle$, $|\chi\rangle$ and $|\omega\rangle$ (orthonormalized)

We construct $|u\rangle = |1, \varphi; 2, \chi; 3, \omega\rangle$

Fermions

$$A|u\rangle = (1/3!)\sum_{\alpha} \epsilon_\alpha P_\alpha |u\rangle \implies \frac{1}{\sqrt{3!}} \begin{vmatrix} |1:\varphi\rangle & |1:\chi\rangle & |1:\omega\rangle \\ |2:\varphi\rangle & |2:\chi\rangle & |2:\omega\rangle \\ |3:\varphi\rangle & |3:\chi\rangle & |3:\omega\rangle \end{vmatrix}$$

Normalize: Replace $\frac{1}{6}$ by $\frac{1}{\sqrt{6}}$

- What if $|\varphi\rangle = |\chi\rangle \implies A|u\rangle = 0$

- What if $|\varphi\rangle = |\chi\rangle = |\omega\rangle \implies A|u\rangle = 0$

How can a symmetric or antisymmetric state be constructed?

General procedure

Consider a system of N identical particles that we can place in the $|\varphi_1\rangle, |\varphi_2\rangle, \ldots, |\varphi_N\rangle$ one-particle ORTHONORMAL states.

1. To each particle we assign a label (a number) and a state, and then in the tensor product state space, $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2 \otimes \ldots \otimes \mathcal{E}_N$, we consider the tensor product of all the states of those N particles:

$$|\psi_{123\ldots N}\rangle = |1{:}\,\varphi_1, 2{:}\,\varphi_2, 3{:}\,\varphi_3, \ldots, N{:}\,\varphi_N\rangle = |1{:}\,\varphi_1\rangle \otimes |2{:}\,\varphi_2\rangle \otimes \ldots \otimes |N{:}\,\varphi_N\rangle$$

2. We construct all the states obtained by **all** possible permutations of the labels assigned to each particle.

For instance:

$$|\psi_{213\ldots N}\rangle = |2{:}\,\varphi_1, 1{:}\,\varphi_2, 3{:}\,\varphi_3, \ldots, N{:}\,\varphi_N\rangle, \qquad |\psi_{231\ldots N}\rangle, \qquad |\psi_{132\ldots N}\rangle, \ldots$$

3a. For bosons → symmetrize:

Add all the obtained states and normalize.

How can a symmetric or antisymmetric state be constructed?

3b. For fermions → antisymmetrize:

Multiply each state by the parity of its permutation (1 for even permutations, -1 for odd permutations), add up all the results and normalize. The final result is equivalent to the so-called Slater determinant:

$$|PAS\rangle = \frac{1}{\sqrt{N!}} \begin{vmatrix} |1:\varphi_1\rangle & |1:\varphi_2\rangle & \cdots & |1:\varphi_N\rangle \\ |2:\varphi_1\rangle & |2:\varphi_2\rangle & \cdots & |2:\varphi_N\rangle \\ \vdots & \vdots & \vdots & \vdots \\ |N:\varphi_1\rangle & |N:\varphi_2\rangle & \cdots & |N:\varphi_N\rangle \end{vmatrix}$$

Consequence: two identical fermions cannot occupy the same one-particle quantum state: notice that if $|\varphi_i\rangle = |\varphi_j\rangle$ then $|PAS\rangle = 0$

This is the generalized Pauli's exclusion principle.

Exchange degeneracy problem removed!!

Let's return to the example of the two spin-1/2 particles.

The infinitely many degenerate states associated with the measurement of S_{1z} and S_{2z}:

$$|\Phi\rangle = \alpha|+,-\rangle + \beta|-,+\rangle$$

transform, after the application of the postulate, into a single state.

For bosons: $$|\Phi\rangle = \frac{1}{\sqrt{2}}[|+,-\rangle + |-,+\rangle]$$

For fermions: $$|\Phi\rangle = \frac{1}{\sqrt{2}}[|+,-\rangle - |-,+\rangle]$$

Note that the two states always enter with the same weight. All other possibilities are not physical because S and A states can not be superposed.

Important remark: Both the triplet and the singlet states are accessible to both fermions and bosons once we work in $\mathcal{E}_r \otimes \mathcal{E}_s$.

Measurements for two identical particles

Let's analyze the consequences of particle indistinguishability (or distinguishability) on the calculation of physical predictions

Performing measurements with two identical particles

Consider a system of two identical particles, one in state $|\varphi\rangle$ and the other one in state $|\chi\rangle$ (which we assume to be orthonormal).

On this system we perform a measurement on each of the two particles, of the same physical magnitude *B* with the associated observables *B*(1) and *B*(2). We will assume that the eigenvalues of the associated operator form a **discrete spectrum and are nondegenerate**

$$B|u_i\rangle = b_i|u_i\rangle$$

We will compute the probability that the measurement of *B* gives b_n for one particle and b_m for the other.

Case a: different results

$n \neq m$ ($b_n \neq b_m, |u_n\rangle \neq |u_m\rangle$, in fact they are orthogonal)

(1) If the particles can be distinguished ("classical") then the state of the system can be taken as $|1:\varphi; 2:\chi\rangle$

and there are two different possibilities for obtaining b_n and b_m in the measurements, namely, $|1:u_n; 2:u_m\rangle$ and $|1:u_m; 2:u_n\rangle$

The sought probability is then the sum of those corresponding to both possibilities:

$$\mathcal{P}_{cl} = |\langle 1:u_n; 2:u_m|1:\varphi; 2:\chi\rangle|^2 + |\langle 1:u_m; 2:u_n|1:\varphi; 2:\chi\rangle|^2 = |\langle u_n|\varphi\rangle\langle u_m|\chi\rangle|^2 + |\langle u_m|\varphi\rangle\langle u_n|\chi\rangle|^2$$

Case a: different results

(2) If the particles cannot be distinguished (identical particles) then we have: (sign + for bosons, sign - for fermions):

- State of the system:

$$|\varphi;\chi\rangle_{\pm} = \frac{1}{\sqrt{2}}(|1:\varphi;2:\chi\rangle \pm |1:\chi;2:\varphi\rangle)$$

- Physical eigenstate associated to the results b_n, b_m:

$$|u_n;u_m\rangle_{\pm} = \frac{1}{\sqrt{2}}(|1:u_n;2:u_m\rangle \pm |1:u_m;2:u_n\rangle)$$

- And the sought probability is the squared modulus of their inner product:

$$\mathcal{P} = \left|{}_{\pm}\langle u_n;u_m|\varphi;\chi\rangle_{\pm}\right|^2$$

Case a: different results

Performing the calculations gives:

$$\mathcal{P}_{cl} = |\langle u_n|\varphi\rangle\langle u_m|\chi\rangle|^2 + |\langle u_m|\varphi\rangle\langle u_n|\chi\rangle|^2$$

$$\mathcal{P}_f = |\langle u_n|\varphi\rangle\langle u_m|\chi\rangle - \langle u_m|\varphi\rangle\langle u_n|\chi\rangle|^2$$

$$\mathcal{P}_b = \underbrace{|\langle u_n|\varphi\rangle\langle u_m|\chi\rangle}_{\text{direct term}} + \underbrace{\langle u_m|\varphi\rangle\langle u_n|\chi\rangle|^2}_{\text{exchange term}}$$

The direct and exchange terms represent two possible paths that the system might have followed from the initial to the final state. These two paths interfere for identical particles (purely quantum effect).

Case a: different results



Schematic representation of the direct term and the exchange term associated with a measurement performed on a system of two identical particles. Before the measurement, one of the particles is known to be in the state $|\varphi\rangle$ and the other one, in the state $|\chi\rangle$. The measurement result obtained corresponds to a situation in which one particle is in the state $|u_n\rangle$ and the other one, in the state $|u_{n'}\rangle$. Two probability amplitudes are associated with such a measurement; they are represented schematically by figures a and b. These amplitudes interfere with a + sign for bosons and with a − sign for fermions.

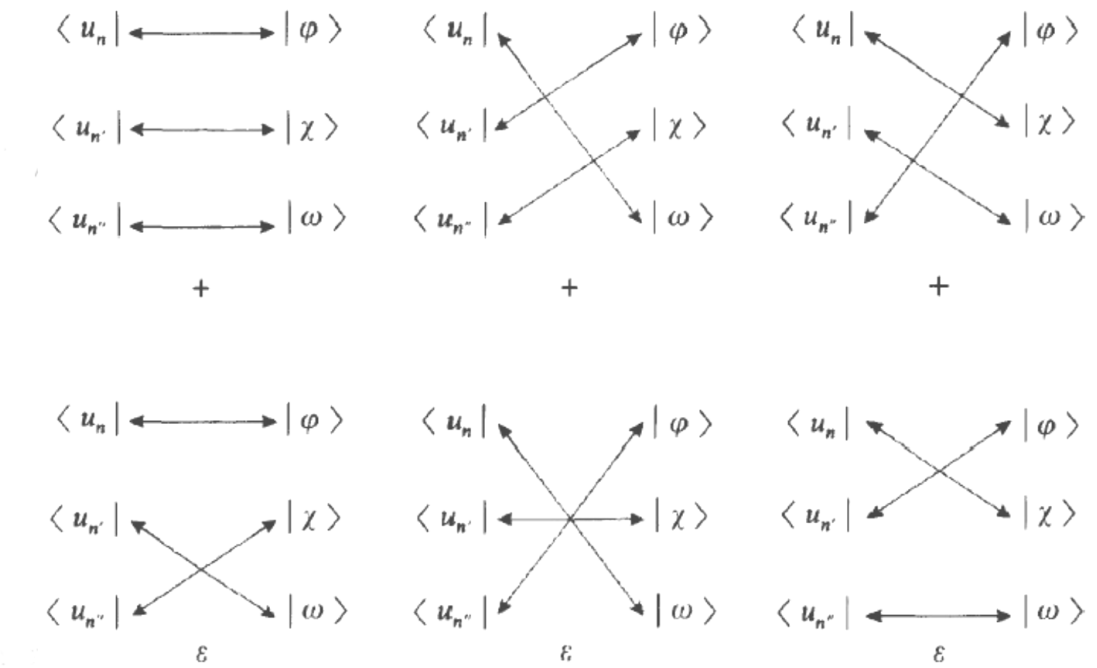
The two kets $|\varphi\rangle$ and $|\chi\rangle$ associated with the initial state can be connected to the two bras $\langle u_n|$ and $\langle u_{n'}|$ associated with the final state by two different "paths".

With each of these paths is associated a probability amplitude, $\langle u_n|\varphi\rangle\langle u_{n'}|\chi\rangle$ or $\langle u_{n'}|\varphi\rangle\langle u_n|\chi\rangle$, and *these two amplitudes interfere with a + sign for bosons and a − sign for fermions.*

Case a: different results

For a system of N identical particles, there will be N! distinct terms

Three identical particles



Schematic representation of the six probability amplitudes associated with a system of three identical particles. Before the measurement, one particle is known to be in the state $|\varphi\rangle$, another, in the state $|\chi\rangle$, and the last one, in the state $|\omega\rangle$. The result obtained corresponds to a situation in which one particle is in the state $|u_n\rangle$, another, in the state $|u_{n'}\rangle$, and the last one, in the state $|u_{n''}\rangle$. The six amplitudes interfere with a sign which is shown beneath each one ($\varepsilon = +1$ for bosons, -1 for fermions).

Case b: same result

$n = m$ $\quad\quad$ $(b_m = b_n, |u_m> = |u_n>)$

(1) Distinguishable:

State of the system: $|1:\varphi, 2:\chi\rangle$;

eigenstate associated to the measurement: $|1:u_n, 2:u_n\rangle$.

(2) Bosons:

State of the system: $|\varphi;\chi\rangle_+$;

eigenstate associated to the measurement: $|u_n,;u_n\rangle_+ = |1:u_n, 2:u_n\rangle$

(3) Fermions:

State of the system: $|\varphi;\chi\rangle_-$;

eigenstate associated to the measurement: $|u_n,;u_n\rangle_- = 0$ (exclusion principle).

Case b: same level

Performing the calculations gives:

$$\mathcal{P}_{cl} = |\langle u_n|\varphi\rangle\langle u_n|\chi\rangle|^2$$

$$\mathcal{P}_f = 0$$

$$\mathcal{P}_b = 2|\langle u_n|\varphi\rangle\langle u_n|\chi\rangle|^2$$

Comparing with the ”classical” result, the probability that the level *n* “is occupied a second time” increases (by a factor of two) for bosons and decreases (to zero) for fermions.

This has important consequences in Quantum Statistical Mechanics.

N independent identical particles

In a system of N identical particles (bosons or fermions) the hamiltonian depends on the variables associated to all of the particles in a special way ⇒ $H(1, \cdots, N)$ must be symmetric with respect to permutations of the particles (it must be the same irrespective of the label assigned to each particle).

Consider a system of independent particles (it is said that *N* particles are **independent** when they do not interact among themselves.), then:

The hamiltonian of a system of independent particles (identical or not) is given by the sum of each particle's hamiltonian:

$$H(1,2,\dots,N) = h(1) + h(2) + \cdots + h(N)$$

For independent **identical** *particles, this implies that all the* $h(j)$ *are the same, but applied to the variables of each particle.*

To evaluate the eigenstates and eigenvalues of the total hamiltonian, *H*, in the total state space $\mathcal{E} = \mathcal{E}_1 \otimes \mathcal{E}_2 \otimes \dots \otimes \mathcal{E}_N$ we simply calculate those of an $h(j)$ in $\mathcal{E}_j$: (*assume that the spectrum of* $h(j)$ *is discrete and non-degenerate*)

$$h(j)|\varphi_n\rangle = e_n|\varphi_n\rangle$$

N independent identical particles

The physical eigenstates of the $H(1, 2, \cdots, N)$ are obtained by symmetrizing the tensor products of N arbitrary individual states $|\varphi_n\rangle$

Bosons

Eigenstates: $$|\Phi_{n_1,n_2,\ldots,n_N}\rangle = SYM[|1:\varphi_{n_1}, 2:\varphi_{n_2}, \ldots, N:\varphi_{n_N}\rangle]$$

Energies: $$E_{n_1,n_2,\ldots,n_N} = e_{n_1} + e_{n_2} + \cdots + e_{n_N}$$

If e_1 is the smallest eigenvalue of $h(j)$ and $|\varphi_1\rangle$ is the associated eigenvector:

Ground state ≡ minimum energy: $E_{1,1,\ldots,1} = Ne_1$ $\quad |\Phi_{1,1,\ldots,1}\rangle = |1:\varphi_1, 2:\varphi_1, \ldots, N:\varphi_1\rangle$

Fermions

Eigenstates: $|\Phi_{n_1,n_2,\ldots,n_N}\rangle = SLATER[|1:\varphi_{n_1}\rangle, |2:\varphi_{n_2}\rangle, \ldots, |N:\varphi_{n_N}\rangle]$ (but only those allowed by the exclusion principle)

Energies: $$E_{n_1,n_2,\ldots,n_N} = e_{n_1} + e_{n_2} + \cdots + e_{n_N}$$

Ground state ≡ minimum energy: The lower single-particle energy levels e_i are occupied consecutively taking into account the exclusion principle.

N independent identical particles

Fermions:

For the ground state, we apply Pauli's exclusion principle.

Arrange the energies in increasing order

$$e_1 < e_2 < e_3 < \cdots < e_n < \cdots$$

Ground state energy of N identical fermions

$$E_{1,2,\ldots N} = e_1 + e_2 + \cdots + e_N$$

$$|\Phi_{1,2,\ldots,N}\rangle = \frac{1}{\sqrt{N!}} \begin{vmatrix} |1:\varphi_1\rangle & |1:\varphi_2\rangle & \cdots & |1:\varphi_N\rangle \\ |2:\varphi_1\rangle & |2:\varphi_2\rangle & \cdots & |2:\varphi_N\rangle \\ \vdots & \vdots & \vdots & \vdots \\ |N:\varphi_1\rangle & |N:\varphi_2\rangle & \cdots & |N:\varphi_N\rangle \end{vmatrix}$$

The highest occupied single-particle level e_N is called the **Fermi energy** (E_F)

Basis in the physical state space

Consider a system of N identical particles. Then: $\mathcal{E} = \mathcal{E}_1 \otimes \ldots \otimes \mathcal{E}_N$

In $\mathcal{E}_i$ we consider a basis $\{|u_i\rangle, i = 1,2, \ldots\}$

So for $\mathcal{E}$ we have a basis $\{|1: u_i; 2: u_j; \ldots; N: u_p\rangle, i, j, \ldots, p = 1,2, \ldots\}$

But we know that the physical state space is NOT $\mathcal{E}$ BUT $\mathcal{E}_S$ or $\mathcal{E}_A$

So, how to find a basis in $\mathcal{E}_S$ or $\mathcal{E}_A$? We can apply the operators S or A to the basis vectors $\{|1: u_i; 2: u_j; \ldots; N: u_p\rangle, i, j, \ldots, p = 1,2, \ldots\}$ of $\mathcal{E}$ and then we can obtain a set of vectors spanning $\mathcal{E}_S$ or $\mathcal{E}_A$.

Let $|\varphi\rangle$ be an arbitrary ket of $\mathcal{E}_S$, for example (the case in which $|\varphi\rangle$ belongs to $\mathcal{E}_A$ can be treated in the same way). $|\varphi\rangle$, which belongs to $\mathcal{E}$, can be expanded in the form:

$$|\varphi\rangle = \sum_{i,j,\ldots,p} a_{i,j,\ldots,p} |1: u_i; 2: u_j; \ldots N: u_p\rangle$$

Since $|\varphi\rangle$, by hypothesis, belongs to $\mathcal{E}_S$, we have $S|\varphi\rangle = |\varphi\rangle$, and we simply apply the operator S to both sides to show that $|\varphi\rangle$ can be expressed in the form of a linear combination of the various kets $S|1: u_i; 2: u_j; \ldots; N: u_p\rangle$.

Now, if we take a specific ket $|1: u_i; 2: u_j; \ldots; N: u_p\rangle$ and consider any permutation out of all possible N! permutations, then after applying S (or A) to this new ket, we always obtain exactly the same ket of $\mathcal{E}_S$ (or $\mathcal{E}_A$).

Basis in the physical state space

So we are led to introduce the idea of occupation number (of an individual state)

For the ket $|1:u_i;2:u_j;\dots;N:u_p\rangle$, the occupation number n_k of the individual state $|u_k\rangle$, is given by the number of times that this state appears in the ket; i.e. the number of particles in the state $|u_k\rangle$. Obviously, $\sum_k n_k = N$

Two different kets with the same occupation numbers give, after the action of the operator S (or A) the **same physical ket** of $\mathcal{E}_S$ (or $\mathcal{E}_A$) which is now denoted as $|n_1, n_2, n_3, \dots\rangle$.

$$|\, n_1, n_2, \dots, n_k, \dots \,\rangle$$
$$= c\, S \,|\, \underbrace{1 : u_1 ; 2 : u_1 ; \dots n_1 : u_1}_{n_1 \text{ particles in the state } |\, u_1 \,\rangle} ; \underbrace{n_1 + 1 : u_2 ; \dots ; n_1 + n_2 : u_2}_{n_2 \text{ particles in the state } |\, u_2 \,\rangle} ; \dots \,\rangle$$

For fermions, S would be replaced by A

Basis in the physical state space

Properties

- The scalar product of two kets $|n_1, n_2, n_3, \dots\rangle$ and $|n_1', n_2', n_3', \dots\rangle$ is different from cero only if all the occupation numbers are the same ($n_k = n_k'$ for all k)

- If the particles are bosons, the sets $|n_1, n_2, n_3, \dots\rangle$ where the n_k are arbitrary (with $\sum_k n_k = N$) form a basis of $\mathcal{E}_S$.

- If the particles are fermions, a basis of $\mathcal{E}_A$ is made by the kets $|n_1, n_2, n_3, \dots\rangle$ in which all the occupation numbers are 1 or 0. (with $\sum_k n_k = N$ and $n_i = 0,1$)